README File Documenting Code and Data for "Regulating Conglomerates: Evidence from an Energy Conservation Program in China"

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Contents

1 Overview **2**

2 Data Availability and Provenance Statement **2**

3 Computational Requirements **5**

4 Description of Programs/Code **6**

5 Instructions for Replicators **6**

1 Overview

- The code in this replication package mainly uses data from four different sources (NBS, 2001-2009, 2011, CMEP, 2001-2010, ATS, 2008-2011, SAIC, 1949-2018)
- The package relies on Stata and Matlab code. The code was run using Stata MP Version 18 and Matlab Version R2023b.
- The package includes two master files: (1) STATA code `Do/Master.do` that outputs all exhibits (main paper, online appendix) from STATA, and (2) MATLAB code `master.m` that outputs three tables (Table 4, Table 5, Table A36) from conducting structural analyses based on estimated data moments. But in terms of execution, (1) `Do/Master.do` also executes `master.m` from within do script.
- Since proprietary data is required to run (1), we provide moment files `Data_Moments/Result/Moment*.dta` which were generated by lines 28-65 of `Do/Master.do` to output final tables and figures.
- The package can be expected to run in about 35-40 minutes.
- Please cite this response when using data and/or code from this replication package.

2 Data Availability and Provenance Statement

Statement about Rights: We certify that the authors of the manuscript have legitimate access to and permission to use the data used in this manuscript.

Summary of data availability: Some data **cannot be made** publicly available.

Details of Raw Data:

This project uses the following data sources:

Table 1: Raw Data Summary

Data file	Source	Notes	Provided
Raw_Data/ASIF/ASIF.dta	NBS(2001-2009,2011)	This data includes all manufacturing firms from the NBS(1998-2009, 2011-2013) files. Data are proprietary and cannot be posted online.	No
Raw_Data/CESD/CESD.dta	CMEP(1998-2010)	This data includes all polluting firms from the CMEP(1998-2010) files. Data are proprietary and cannot be posted online.	No
Raw_Data/ATS/ATS.dta	ATS(2008-2011)	This data includes all firms from the ATS(2008-2011) files. Data are proprietary and cannot be posted online.	No
Raw_Data/CARD/CARD.dta	SAIC(1949-2018)	This data includes registration information for all firms from SAIC(1949-2018) files. Data are proprietary and cannot be posted online.	No
Raw_Data/Census/Census_2004.dta	NBS(2004)	This data includes all firms from the NBS(2004) file. Data are proprietary and cannot be posted online.	No
Raw_Data/Patent/Patent.dta	Incopat(2001-2010)	This data includes patent information for all firms from the Incopat(2001-2010) files. Data are proprietary and cannot be posted online.	No
Raw_Data/Top1000/Top1000_Energy_Saving_Target.dta	NDRC(2008-2010)	This data includes energy saving targets for all Top1,000 firms evaluated by the government from the NDRC(2008-2010) files.	No
Raw_Data/Top1000/Top1000_Industry.dta	NDRC,NBS(2007)	This data includes energy consumption for all Top1,000 firms reported in the bulletin on Top 1,000 firms (2007).	No
Raw_Data/Top1000/Top1000_Compliance(_Noncompliance).dta	NDRC(2006-2010)	This data includes compliance information for all Top1,000 firms from the NDRC(2006-2010) files.	No
Raw_Data/Top10000/Top10000_QXB_ID.dta	NDRC(2012)	This data includes name list of all Top10,000 firms from the NDRC(2012) file.	No
Raw_Data/Top10000/Top10000_Energy_Saving_Target(_Legcode/_Name).dta	NDRC(2012)	This data includes energy saving targets for all Top10,000 firms from the NDRC(2012) file.	No
Raw_Data/Key_Monitor/gas2007_legcode(name).dta	CMEP (2007)	This data includes all monitored air-polluting firms from the CMEP (2007) file.	No
Raw_Data/Key_Monitor/water2007_legcode(name).dta	CMEP (2007)	This data includes all monitored water-polluting firms from the CMEP (2007) file.	No
Raw_Data/Industry/Industry_Energy_2001_2011.dta	CESY(2001-2011)	This data includes energy consumption for all 2-digit manufacturing industries from the China Energy Statistical Yearbook (CESY, 2001-2011) file.	No
Raw_Data/Industry/Industry_Energy_2004.dta	CESY(2004)	This data includes energy consumption for all 2-digit manufacturing industries from the CESY(2004) file.	No
Raw_Data/Industry/Industry_Energy_Use_2010.dta	CESY(2010)	This data includes energy consumption for all 2-digit manufacturing industries from the CESY(2010) file.	No
Raw_Data/Industry/Industry_Output_2001_2011.dta	CISY(2001-2011)	This data includes output for all 2-digit manufacturing industries from the China Industry Statistical Yearbook. (CISY, 2001-2011) file.	No
Raw_Data/Industry/Industry_Control_2009_2013.dta	CESY,CISY(2009-2013)	This data includes all 2-digit manufacturing industries from the CESY(2009-2013) and CISY(2009-2013) files.	No
Raw_Data/Industry/Nine_Industry_Energy_Mix.dta	CESY(2001-2010)	This data includes energy mix for nine energy-intensive industries from the CESY (2001-2010).	No
Raw_Data/Density/City(Province)_Density.dta	CSMAR(2006)	This data includes air pollution density of all cities(provinces) from the CSMAR(2006) file.	No
Raw_Data/Density/City_Population.dta	CCSY(2006)	This data includes population of all cities from China City Statistical Yearbook(CCSY,2006).	No

Table 2: Subsamples and Processed Data of Proprietary Raw Data That We Provide

Data file	Source	Notes	Provided
Raw_Data/ASIF/ASIF_Top1000_Top10000.dta	NBS(2001-2009,2011)	This data includes Top1,000 and Top10,000 firm samples from the NBS(1998-2009, 2011-2013) files. Subsample of Raw_Data/ASIF/ASIF.dta	No
Raw_Data/CESD/CESD_Top1000_Top10000.dta	CMEP(1998-2010)	This data includes Top1,000 and Top10,000 firm samples from the CMEP(1998-2010) files. Subsample of Raw_Data/CESD/CESD.dta. Firms in industries that rely mainly on electricity (by over 30%) and thermal power generation are excluded.	No
Raw_Data/CESD/CESD_Top1000_Top10000_Raw.dta	CMEP(1998-2010)	This data includes Top1,000 and Top10,000 firm samples from the CMEP(1998-2010) files. Subsample of Raw_Data/CESD/CESD.dta.	No
Raw_Data/CESD/Energy_Type.dta	CMEP(1998-2010) ATS(2008-2011)	This data includes main energy types used by each mannufaturing industry from the CMEP(1998-2010) files. Processed data from Raw_Data/CESD/CESD.dta and Raw_Data/ATS/ATS.dta.	No
Raw_Data/CESD/Industry_Electricity_Share.dta	CMEP(1998-2010) ATS(2008-2011)	This data includes electricity reliance of each mannufaturing industry from the CMEP(1998-2010) files. Processed data from Raw_Data/CESD/CESD.dta and Raw_Data/ATS/ATS.dta.	No
Raw_Data/ATS/ATS_Supplement(_Coal/_Electricity/_Energy/_Other)_Legcode(Name).dta	ATS(2008-2011)	This data includes financial information and energy consumption for all firms from the ATS(2008-2011) files. Processed data from Raw_Data/ATS/ATS.dta.	No
Raw_Data/ATS/ATS_Top1000_Top10000.dta	NBS(2001-2008) ATS(2008-2011)	This data includes Top1,000 and Top10,000 firm samples from NBS(2001-2008) and ATS(2008-2011) files. Firms that didn't report original value of fixed assets in 2008 or reported different values (by over 25%) in two datasets are excluded. Processed data from Raw_Data/ASIF/ASIF.dta and Raw_Data/ATS/ATS.dta.	No
Raw_Data/CARD/related_firms(_legcode/_name).dta	SAIC(1949-2018)	This data includes all related manufacturing firms above designated size of Top1,000 firms from SAIC(1949-2018) files. Processed data from Raw_Data/CARD/CARD.dta. Related firms by up to two levels of investment relations and at least 25% ownership requirement.	No
Raw_Data/CARD/related_firms(_legcode/_name)_1-3.dta	SAIC(1949-2018)	This data includes all related manufacturing firms above designated size of Top1,000 firms from SAIC(1949-2018) files. Processed data from Raw_Data/CARD/CARD.dta. Related firms by up to six(two) levels of investment relations and at least 20%(50%) ownership requirement.	No
Raw_Data/CARD/Related_Firms.dta	SAIC(1949-2018)	This data includes all related pairs for each Top1,000 firm and their related firms from SAIC(1949-2018) files. Processed data from Raw_Data/CARD/CARD.dta. Related firms by up to two levels of investment relations and at least 25% ownership requirement.	No
Raw_Data/CARD/Related_Robust.dta	SAIC(1949-2018)	This data includes all related firms for Top1,000 firms from SAIC(1949-2018) files. Processed data from Raw_Data/CARD/CARD.dta.	No
Raw_Data/CARD/Conglomerate_ID_Related_Legcode(_Name).dta	SAIC(1949-2018)	This data includes conglomerate information for related firms of Top1,000 firms from SAIC(1949-2018) files. Processed data from Raw_Data/CARD/CARD.dta.	No
Raw_Data/CARD/Conglomerate_ID_Top1000_Legcode(_Name/_Name2).dta	SAIC(1949-2018)	This data includes conglomerate information for Top1,000 firms from SAIC(1949-2018) files. Processed data from Raw_Data/CARD/CARD.dta.	No
Raw_Data/CARD/Related_Legcode.dta	IntSig Information(2018)	This data includes legal code for all related firms of Top1,000 firms from IntSig Information(2018). Processed data from Raw_Data/CARD/CARD.dta.	No
Raw_Data/CARD/Related_Capital.dta	SAIC(1949-2018)	This data includes capital information for all related firms of Top1,000 firms from SAIC(1949-2018) files. Processed data from Raw_Data/CARD/CARD.dta.	No
Raw_Data/CARD/Top10000_Related_Capital.dta	SAIC(1949-2018)	This data includes capital information for all related firms of Top10,000 firms from SAIC(1949-2018) files. Processed data from Raw_Data/CARD/CARD.dta.	No
Raw_Data/CARD/Shareholding_Change.dta	IntSig Information(2018)	This data includes all related firms of Top1,000 firms that experienced significant ownership changes between 2007 and 2018. Processed data from Raw_Data/CARD/CARD.dta.	No
Raw_Data/Patent/Patent_Top1000_Top10000.dta	Incopat(2001-2010)	This data includes Top1,000 and Top10,000 firm samples from the Incopat(2001-2010) files. Processed data from Raw_Data/Patent/Patent.dta.	No
Raw_Data/Top1000/Top1000_QXB_ID.dta	IntSig Information(2018)	This data includes corresponding legal code and name of Top1,000 firms from the IntSig Information(2018). Processed data from Raw_Data/CARD/CARD.dta.	No
Raw_Data/Top1000/Top1000_Capital.dta	IntSig Information(2018)	This data includes capital information for all Top1,000 firms from IntSig Information(2018). Processed data from Raw_Data/CARD/CARD.dta.	No
Raw_Data/Top10000/fudan_affiliates.xlsx	SAIC(1949-2018)	This data includes ID for affiliates of Top10,000 firms from the SAIC(1949-2018) files. Processed data from Raw_Data/CARD/CARD.dta.	No
Raw_Data/Top10000/fudan_investment_new.xlsx	SAIC(1949-2018)	This data includes ID for investment of Top10,000 firms from the SAIC(1949-2018) files. Processed data from Raw_Data/CARD/CARD.dta.	No
Raw_Data/Top10000/fudan_shareholder_new.xlsx	SAIC(1949-2018)	This data includes ID for shareholders of Top10,000 firms from the SAIC(1949-2018) files. Processed data from Raw_Data/CARD/CARD.dta.	No
Raw_Data/Top10000/fudan_shareholder_all.xlsx	SAIC(1949-2018)	This data includes name for shareholders of Top10,000 firms from the SAIC(1949-2018) files. Processed data from Raw_Data/CARD/CARD.dta.	No
Raw_Data/Top10,000/fudan_shareholder_investment_all_new.xlsx	SAIC(1949-2018)	This data includes ID for firm shareholder investment of Top10000 firms from the SAIC(1949-2018) files. Processed data from Raw_Data/CARD/CARD.dta.	No
Raw_Data/Top10000/fudan_personshareholder_investment_new.xlsx	SAIC(1949-2018)	This data includes ID for peronal shareholder investment of Top10,000 firms from the SAIC(1949-2018) files. Processed data from Raw_Data/CARD/CARD.dta.	No
Raw_Data/Top10000/Government_Shareholder.dta	SAIC(1949-2018)	This data includes list of government shareholders for Top10,000 firm from the SAIC(1949-2018) files. Processed data from Raw_Data/CARD/CARD.dta.	No

All of these files are in `dta` format, which include variable names and labels. The proprietary data were purchased by Fudan University. For the purpose of replication, researchers can contact the data center director at Fudan University, Yiyi Xin at `econdata@fudan.edu.cn`. Additionally, authors will assist with any reasonable replication attempts for two years following publication. Please contact Qiaoyi Chen at `joychen@fudan.edu.cn`.

Details of Data Used in Analyses:

We combine data from NBS(1998-2009, 2011-2013), CMEP(1998-2010), ATS(2008-2011), and SAIC(1949-2018) to generate analyse files which can be shared publicly. These files are sufficient to estimate most of our analyses and to produce the main results of the paper. A few of our analyses also use public data from official documents or from different kinds of statistical yearbooks in China.

3 Computational Requirements

All analysis were conducted in Stata MP Version 18 and Matlab Version R2023b. Code for this project was last run on 2021 Macbook M1 Pro (Apple Silicon Chip; 10-core CPU with 32 GB of memory) Runtime was 35 minutes. The files `Log_Stata.smcl` report the logs from a run on February 10, 2024.

Also, the following MATLAB Toolbox add-ons are required to run the program.

- Statistical and Machine Learning Toolbox
- Optimization Toolbox

Finally, the following STATA packages are required to run the program. Lines 1-10 in `Master.do` executes the installation of these packages: (also stored in `StataAdoBackup/`)

- addplot
- ebalance
- listtex
- unique
- binscatter
- ftools
- estout
- winsor2
- reghdfe
- fdta
- parmest
- xfill
- texdoc

4 Description of Programs/Code

Code is located in four directories:

- `Data_Generation_Program`: This folder contains routines to clean data and produce analysis files that are saved in `Data` and `Data_Moments`. Code in this folder relies on proprietary data.
- `Do`: This folder contains routines to produce tables, graphs and scalars that are saved in `Table`, `Figure`, `Appendix_Table`, `Appendix_Figure`, and `Scalars` based on analysis data. Code in this folder relies on data in the folder `Data`.
- `Data_Moments/Do`: This folder contains routines to produce data moments for structural estimation that are saved in `Data_Moments/Result`. Code in this folder relies on data in the folder `Data`.
- `Structural_Analysis`: This folder contains routines to estimate structural analyses and outputs are saved in `Structural_Analysis/Output`. Code in this folder relies on estimated data moments in the folder `Data_Moments/Result`.

Master files. The following master files run all of the analyses in the paper:

1. Data Cleaning in Stata (lines 31-70). `Do/Master.do`: Runs routines to clean data files. This file calls the scripts listed in Table 3. This script relies on proprietary data. Replicators with access to proprietary data can find the data in `Raw_Data`.
2. Structural Estimation in Matlab. These routines can be called within Stata in lines 72-75 of `Do/Master.do`. Alternatively, the file `Structural_Analysis/master.m` can be used to run the structural estimation routines. This file calls the scripts listed in Table 4. This script relies on estimated data moments provided in this archive and located in the folder `Data_Moments/Result`.
3. Graphs and Tables in Stata (lines 78-168). `Do/Master.do`: Generates graphs and tables. This file calls the scripts listed in Table 6. This script relies on analyses data files provided in this archive and located in `Data` and (public) raw data in `Raw_Data`.

Final Outputs are deposited in the following directories:

- Figures and tables are saved in `Figure`, `Table`, `Appendix_Figure`, and `Appendix_Table`.
- Scalars are saved in `Scalars`.
- Outputs from structural estimation are saved in `Structural_Analysis/Output`.

5 Instructions for Replicators

The following is the way to replicate the results of the paper:

- Use the following STATA script to run two master files: `Do/Master.do`. Prior to running this program, replicators must set working directories in the following locations:
    1. Line 16 of `Do/Master.do`

2. Lines 5 and 6 of `Structural_Analysis/define_directory.m` . Please note the structure of the paths in these lines. Specifically, the new paths should look like `PATH/Structural_Analy` where `PATH` is your path.

- Also, since MATLAB master script `Structural_Analysis/master.m` is run and executed from within `Do/Master.do`, it is essential for the replicator to specify the exact MATLAB executable in Line 21 of `Do/Master.do`; otherwise not only `master.m` but also `Table2.do`, `TableA24.do`, and `TableA38.do` will not run properly.

  1. For most cases, specifying Line 21 as follows will work: `global matlab "!matlab"`
  2. For cases where the following errors are raised (`zsh commmand not found:matlab`), it means that UNIX/Terminal could not locate MATLAB executable. In this case, it is important for replicator to specify the path of matlab executable. For MacOS users with MATLAB R_2023b, for example, Line 21 should be as follows:
     `global matlab ‘‘!/Applications/MATLAB_R2023b.app/bin/matlab’’`

  When setting working directories please be mindful of noting when a directory ends with a forward slash (i.e., "/") and when it does not, as this may impede the code from running correctly.

Known Issues

- `Table2.do`, `TableA24.do`, and `TableA38.do` has lines of code that runs MATLAB code within the do script. For example, Lines 100, 322, 549 of `Table2.do` runs MATLAB codes `Distance_Matching_Output.m`, `Distance_Matching_Placebo.m`, `Distance_Matching_Energy.m`, respectively. Depending on your MATLAB configuration (see this link), your STATA may not find matlab executable and output the following: `zsh command not found: matlab`. In order to correct for this problem, specify Line 21 of `Do/Master.do` to specify MATLAB executable path for the OS: `!/Applications/MATLAB_R2023b.app/bin/matlab` if your MATLAB version is MATLAB R_2023b. If error `command not found: matlab` is not raised from Line 22 of `Do/Master.do`, the rest of the package will run successfully.

Table 3: Cleaning Code

Code	Purpose of Code
Data_Generation_Program/Data_for_Table1.do	Prepares Data for Table Table 1, A5, A7 Panel A, A10, A11, A12, A13, A15, A16, A17, A18, A19, A20, A21, A32, A34 and Figure 3, A6, A9, A10, A14, A19, saves it in Data
Data_Generation_Program/Data_for_Table2.do	Prepares Data for Table 2, A25, A27, A28, A29, A35 and Figure 4, A16, A18, A20, saves it in Data
Data_Generation_Program/Data_for_Table3.do	Prepares Data for Table 3, A21, A30 and Figure 2 Panel B, 5, saves it in Data
Data_Generation_Program/Data_for_TableA1.do	Prepares Data for Table A1, saves it in Data
Data_Generation_Program/Data_for_TableA2.do	Prepares Data for Table A2, saves it in Data
Data_Generation_Program/Data_for_TableA3.do	Prepares Data for Table A3, saves it in Data
Data_Generation_Program/Data_for_TableA4.do	Prepares Data for Table A4, saves it in Data
Data_Generation_Program/Data_for_TableA6.do	Prepares Data for Table A6, saves it in Data
Data_Generation_Program/Data_for_TableA7.do	Prepares Data for Table A7 Panel B, saves it in Data
Data_Generation_Program/Data_for_TableA8.do	Prepares Data for Table A8, A9 Panel A and Figure A8, saves it in Data
Data_Generation_Program/Data_for_TableA9.do	Prepares Data for Table A9 Panel B, saves it in Data
Data_Generation_Program/Data_for_TableA14.do	Prepares Data for Table A14 and Figure A11, saves it in Data
Data_Generation_Program/Data_for_TableA22.do	Prepares Data for Table A22 and Figure A15, saves it in Data
Data_Generation_Program/Data_for_TableA23.do	Prepares Data for Table A23, saves it in Data
Data_Generation_Program/Data_for_TableA24.do	Prepares Data for Table A24, saves it in Data
Data_Generation_Program/Data_for_TableA26.do	Prepares Data for Table A26 and Figure A17, saves it in Data
Data_Generation_Program/Data_for_TableA37.do	Prepares Data for Table A37 and Figure A22, saves it in Data
Data_Generation_Program/Data_for_TableA38.do	Prepares Data for Table A38, saves it in Data
Data_Generation_Program/Data_for_FigureA5.do	Prepares Data for Figure A5, saves it in Data
Data_Generation_Program/Data_for_FigureA12.do	Prepares Data for Figure A12, saves it in Data
Data_Generation_Program/Data_for_FigureA13.do	Prepares Data for Figure A13, saves it in Data
Data_Generation_Program/Data_for_Moment1.do	Prepares Data for Moment 1, saves it in Data
Data_Generation_Program/Data_for_Moment3.do	Prepares Data for Moment 3 and Figure 2 Panel A, saves it in Data
Data_Generation_Program/Data_for_Moment4.do	Prepares Data for Moment 4, saves it in Data
Data_Generation_Program/Data_for_Moment5.do	Prepares Data for Moment 5, saves it in Data_Moments/Data
Data_Generation_Program/Data_for_Scalars.do	Prepares Data for Scalars, saves it in Data

Table 4: Structural Estimation Code

Type of Analysis	Code Use
1. Structural Moments	Outputs are saved in Data_Moments/Result
This file generates moments for firm size distribution.	Data_Moments/Do/Moment1.do
This file generates moments for within-conglomerate distribution.	Data_Moments/Do/Moment2.do
This file generates moments for share of small firms.	Data_Moments/Do/Moment3.do
This file generates moments for cost share of variable inputs	Data_Moments/Do/Moment4.do
This file generates moments for energy share in variable inputs	Data_Moments/Do/Moment5.do
2. Structural Estimation	Structural_Analysis/master.m
This file runs the following structural analysis:	
2.1 Baseline Estimation	Located
Estimate parameters, output Table 4	Structural_Analysis/Estimation/Main.m
Compare model fits, output Figure 6	Structural_Analysis/Estimation/Main_test.m
2.2 Estimation: Robustness	Located
Alternative combinations ρ and σ	Structural_Analysis/Estimation Menu/Main.m
Imperfect substitution	Structural_Analysis/Estimation ImpSub/Main.m
	Structural_Analysis/Estimation ImpSub/Main_ste.m
2.3 Sensitivity Analysis	Located
Outputs TableA36	Structural_Analysis/Sensitivity Analysis/Main_ste.m
	Structural_Analysis/Sensitivity Analysis/Main_ste_menu.m
2.4 Counterfactuals	Located
Output Table 5	Structural_Analysis/Counterfactual Baseline/Main.m
Output Figure 7	Structural_Analysis/Counterfactual Baseline/Main_compare.m
	Structural_Analysis/Counterfactual Robustness/Main_compare.m
2.5 Robustness: Energy Efficiency	Located
Output Figure A2 Panel A	Structural_Analysis/Energy Efficiency/Main.m
2.6 Robustness: Heterogeneous Energy Efficiency	Located
Output Figure A2 Panel B	Structural_Analysis/Heterogeneous EE/Main.m
2.7 Robustness: Imperfect Substitution	Located
Output Figure A2 Panel C	Structural_Analysis/Imperfect Sub/Main.m

Note: †To limit computing time, the estimation commands for robustness checks discussed in the Online Appendix are commented out.

Table 5: Result of Structural Estimation that is taken as input by STATA

Estimation Result File in `Structural_Analysis/`	STATA `.do` script that uses the result	Output Exhibit
`Estimation/moments.txt`	`Do/Figure6.do`	Figure 6 Panel A
`Counterfactual Baseline/results_figure_6B.txt`	`Do/Figure6.do`	Figure 6 Panel B
`Counterfactual Baseline/results.txt`	`Do/Figure7.do`	Figure 7 Panel A, B
`Energy Efficiency/results.txt`	`Do/FigureA2.do`	Figure A2 Panel A
`Heterogeneous EE/results.txt`	`Do/FigureA2.do`	Figure A2 Panel B
`Imperfect Sub/results.txt`	`Do/FigureA2.do`	Figure A2 Panel C
`Counterfactual Robustness/results_85_75.txt`	`Do/FigureA2.do`	Figure A2 Panel C
`Counterfactual Robustness/results_90_70.txt`	`Do/FigureA2.do`	Figure A2 Panel C
`Counterfactual Robustness/results_90_90.txt`	`Do/FigureA2.do`	Figure A2 Panel C
`Counterfactual Robustness/results_95_75.txt`	`Do/FigureA2.do`	Figure A2 Panel C
`Counterfactual AdjK/results.txt`	`Do/FigureA2.do`	Figure A2 Panel C
`No Conglomerate/results.txt`	`Do/FigureA2.do`	Figure A2 Panel C
`Counterfactual Baseline/results.txt`	`Do/FigureA2.do`	Figure A2 Panel C
`Counterfactual Baseline/results_graphs.txt`	`Do/FigureA3.do`	Figure A3 Panel A, B
`Heterogeneous EE/results_robust.txt`	`Do/FigureA21.do`	Figure A21

Table 6: Code for Graphs and Tables[†]

Code	Purpose of Code
Do/Figure1.do	Saves Figure 1 in Figure
Do/Figure2.do	Saves Figure 2 in Figure
Do/Figure3.do	Saves Figure 3 in Figure
Do/Figure4.do	Saves Figure 4 in Figure
Do/Figure5.do	Saves Figure 5 in Figure
Do/Figure6.do	Saves Figure 6 in Figure
Do/Figure7.do	Saves Figure 7 in Figure
Do/Table1.do	Saves Table 1 in Table
Do/Table2.do	Saves Table 2 in Table
Do/Table3.do	Saves Table 3 in Table
Structural_Analysis/Output/Table4.m	Saves Table 4 in Structural_Analysis/Output/Table4.txt
Structural_Analysis/Output/Table5.m	Saves Table 5 in Structural_Analysis/Output/Table5.txt
Do/FigureA2.do	Saves Figure A2 in Appendix_Figure
Do/FigureA3.do	Saves Figure A3 in Appendix_Figure
Do/FigureA5.do	Saves Figure A5 in Appendix_Figure
Do/FigureA6.do	Saves Figure A6 in Appendix_Figure
Do/FigureA8.do	Saves Figure A8 in Appendix_Figure
Do/FigureA9.do	Saves Figure A9 in Appendix_Figure
Do/FigureA10.do	Saves Figure A10 in Appendix_Figure
Do/FigureA11.do	Saves Figure A11 in Appendix_Figure
Do/FigureA12.do	Saves Figure A12 in Appendix_Figure
Do/FigureA13.do	Saves Figure A13 in Appendix_Figure
Do/FigureA14.do	Saves Figure A14 in Appendix_Figure
Do/FigureA15.do	Saves Figure A15 in Appendix_Figure
Do/FigureA16.do	Saves Figure A16 in Appendix_Figure
Do/FigureA17.do	Saves Figure A17 in Appendix_Figure
Do/FigureA18.do	Saves Figure A18 in Appendix_Figure
Do/FigureA19.do	Saves Figure A19 in Appendix_Figure
Do/FigureA20.do	Saves Figure A20 in Appendix_Figure
Do/FigureA21.do	Saves Figure A21 in Appendix_Figure
Do/FigureA22.do	Saves Figure A22 in Appendix_Figure
Do/TableA1.do	Saves Table A1 in Appendix_Table
Do/TableA2.do	Saves Table A2 in Appendix_Table
Do/TableA3.do	Saves Table A3 in Appendix_Table
Do/TableA4.do	Saves Table A4 in Appendix_Table
Do/TableA5.do	Saves Table A5 in Appendix_Table
Do/TableA6.do	Saves Table A6 in Appendix_Table
Do/TableA7.do	Saves Table A7 in Appendix_Table
Do/TableA8.do	Saves Table A8 in Appendix_Table
Do/TableA9.do	Saves Table A9 in Appendix_Table
Do/TableA10.do	Saves Table A10 in Appendix_Table
Do/TableA11.do	Saves Table A11 in Appendix_Table
Do/TableA12.do	Saves Table A12 in Appendix_Table
Do/TableA13.do	Saves Table A13 in Appendix_Table
Do/TableA14.do	Saves Table A14 in Appendix_Table
Do/TableA15.do	Saves Table A15 in Appendix_Table
Do/TableA16.do	Saves Table A16 in Appendix_Table
Do/TableA17.do	Saves Table A17 in Appendix_Table
Do/TableA18.do	Saves Table A18 in Appendix_Table
Do/TableA19.do	Saves Table A19 in Appendix_Table
Do/TableA20.do	Saves Table A20 in Appendix_Table
Do/TableA21.do	Saves Table A21 in Appendix_Table
Do/TableA22.do	Saves Table A22 in Appendix_Table
Do/TableA23.do	Saves Table A23 in Appendix_Table
Do/TableA24.do	Saves Table A24 in Appendix_Table
Do/TableA25.do	Saves Table A25 in Appendix_Table
Do/TableA26.do	Saves Table A26 in Appendix_Table
Do/TableA27.do	Saves Table A27 in Appendix_Table
Do/TableA28.do	Saves Table A28 in Appendix_Table
Do/TableA29.do	Saves Table A29 in Appendix_Table
Do/TableA30.do	Saves Table A30 in Appendix_Table
Do/TableA31.do	Saves Table A31 in Appendix_Table
Do/TableA32.do	Saves Table A32 in Appendix_Table
Do/TableA33.do	Saves Table A33 in Appendix_Table
Do/TableA34.do	Saves Table A34 in Appendix_Table
Do/TableA35.do	Saves Table A35 in Appendix_Table
Structural_Analysis/Output/TableA36.m	Saves Table A36 in Structural_Analysis/Output/TableA36.txt
Do/TableA37.do	Saves Table A37 in Appendix_Table
Do/TableA38.do	Saves Table A38 in Appendix_Table

Notes: [†] Figure A1 and A7 in the paper are examples of related firms which can be plotted directly based on CARD data. Figure A4 is a record for interviews conducted by the authors.
Table 4, 5 and A36 are produced by MATLAB routines. Figure 6, 7, A2, A3, and A21 are produced based on Matlab results. Table A36 is calculations based on the literature and the world bank report.
See Table **??** for additional figures and tables produced in MATLAB.

References

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